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Indian Standard

SAFETY OF HOUSEHOLD AND SIMILAR ELECTRICAL APPLIANCES — SPECIFICATION

PART 2 PARTICULAR REQUIREMENTS

Section 12 Warming Plates and Similar Appliances

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FOREWORD

This Indian Standard was adopted by the Bureau of Indian Standards, after the draft finalized by the Electrical Appliances Sectional Committee, had been approved by the Electrotechnical Division Council.

Electric warming plates, warming trays and similar devices are used for keeping the food warm without the risk of its getting burnt.

The standard covers the general and safety requirements.

This standard does not cover the performance requirements which is covered under a separate composite standard IS 13231 : 1991 (*under revision*).

This standard is to be read in conjunction with IS 302-1 (1979), 'Safety of household and similar electrical appliances : Part 1 General requirements (*fifth revision*)'. For the sake of convenience, the clauses of this standard correspond to those of IS 302-1 (1979) which are applicable which means that relevant provisions of the clause shall apply, or not applicable and the sub-clauses or portions thereof which are not applicable are indicated as under:

- a) In case of a clause where it is applicable or not applicable, the wording used in this clause of IS 302-1 (1979) is applicable/not applicable.
- b) In case of a sub-clause or part thereof not applicable.

Wherever a sub-clause of IS 302-1 (1979) is to be replaced by a new test, it has been indicated as under:

Replacement — followed by the new text.

Any addition to the existing provisions of a sub-clause of IS 302-1 (1979) has been indicated as under.

Addition — followed by the text of the additional matter.

Clauses/tables which are additional to those of IS 302-1 (1979) are numbered starting from 101 and additional sub-clauses are numbered with the main clauses number followed by 101, 102, etc, for example, 7.101.

Additional annexes have been numbered starting from AA.

Should however any deviation exist between IS 302-1 (1979) and this standard, the provisions of the latter shall apply.

While preparing this standard, assistance has been derived from the following publications:

IEC Pub 335-1 (1976) Safety of household and similar electrical appliances : Part 1 General requirements, issued by the International Electrotechnical Commission.

IEC Publication 335-2-12 (1987) Safety of household and similar electric appliances : Part 2 Particular requirements for warming plates and similar appliances, issued by the International Electrotechnical Commission.

The Indian Standards which are necessary adjuncts to this standard are given in Annex BB.

For the purpose of deciding whether a particular requirement of this standard is complied with, the final value, observed or calculated, expressing the result of a test or analysis, shall be rounded off in accordance with IS 2 : 1960 'Rules for rounding off numerical values (*revised*)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

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SAFETY OF HOUSEHOLD AND SIMILAR ELECTRICAL APPLIANCES — SPECIFICATION

PART 2 PARTICULAR REQUIREMENTS

Section 12 Warming Plates and Similar Appliances

1 SCOPE

This clause of IS 302-1 (1979) is applicable except as follows:

1.1 Replacement

This standard applies to electric warming plates, warming trays and similar appliances intended for keeping food or vessels warm, for household and similar purposes, their rated voltage not exceeding 250 V single phase ac or dc.

Appliances not intended for normal household use, but which nevertheless may be a source of danger to the public, such as appliances intended to be used by laymen in shops, in light industry and on farms are also within the scope of this standard.

This standard does not take into account the special hazards which exist in nurseries and other places where there are young children or aged or infirm persons without supervision in such cases, additional requirements may be necessary.

This standard does not apply to:

- appliances made of flexible material, such as textile material.

- appliances intended to be used in locations where special conditions prevail, such as the presence of a corrosive or explosive atmosphere (dust, vapour or gas).

- appliances designed exclusively for commercial catering or industrial purposes.

Until an Indian Standard for appliances for commercial catering purposes is issued, this standard may form a guide to suitable requirements and tests for such appliances.

For appliances intended to be used in vehicles or on board ships or aircraft, additional requirements may be necessary.

2 DEFINITIONS

This clause of IS 302-1 (1979) is applicable except as follows:

2.29 Replacement

Conditions of adequate heat discharge denote that the appliance is operated under the following conditions.

Appliances provided with a thermostat are operated with or without a vessel on the heated surface, whichever is the more unfavourable, the vessel being a shallow pan, 15 cm in diameter, which is filled with water to a height of at least 25 mm, unless vessels are delivered with the appliance or specified in the manufacturer's instructions, in which case these vessels are used.

Other appliances are operated without a vessel or other object on the heated surface.

Additional Definitions:

2.101 Warming Plate

An appliance having a heated surface on which food or vessels can be placed, the appliance being intended to keep the food or the vessels at a temperature suitable for serving or to bring the food or the vessels to that temperature.

2.102 Warming Tray

A warming plate so designed that it can receive a variety of different vessels.

2.103 Particularized Warming Plate

A warming plate so designed that it can receive only particular food-serving or beverage-serving vessels; these vessels may either be delivered with the warming plate or be specified in the manufacturer's instructions.

3 GENERAL REQUIREMENTS

This clause of IS 302-1 (1979) is applicable.

4 GENERAL NOTES ON TESTS

This clause of IS 302-1 (1979) is applicable except as follows:

4.2 Addition

For appliances intended to be partially or completely immersed in water for cleaning,

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three additional samples are required for the tests of **15.101**.

Additional Sub-clause

4.101 The appliances are tested as portable appliances, unless it is clear from their design and the manufacturer's that they are intended to be fixed to a supporting surface.

5 RATING

This clause of IS 302-1 (1979) is applicable.

6 CLASSIFICATION

This clause of IS 302-1 (1979) is applicable except as follows:

6.1 Addition

This standard recognizes appliances which are intended to be partially or completely immersed in water for cleaning; it does not recognize watertight appliances.

7 MARKING

The clause of IS 302-1 (1979) is applicable except as follows:

7.12 Addition

Appliances intended to be partially or completely immersed in water for cleaning shall be accompanied by an instruction sheet stating that the connector must be removed before the appliance is cleaned and that the appliance inlet must be dried before the appliance is used again.

For appliance intended to be used with a connector incorporating a thermostat, the instruction sheet shall state that such appliances must only be used with the appropriate connector.

Appliances having heated surfaces of glass, ceramic or similar brittle material shall be accompanied by an instruction sheet which includes the substance of the following:

Warning — Do not use the appliance if the heated surface is cracked.

Additional Sub-clause

7.101 The appliances may also be marked with the Standard Mark.

8 PROTECTION AGAINST ELECTRIC SHOCK

The clause of IS 302-1 (1979) is applicable except as follows:

8.7 Not applicable.

9 STARTING OF MOTOR-OPERATED APPLIANCES

This clause of IS 302-1 (1979) is not applicable.

10 INPUT AND CURRENT

This clause of IS 302-1 (1979) is applicable.

11 TEMPERATURE RISE

This clause of IS 302-1 (1979) is applicable except as follows:

11.2 Replacement

Appliances for building in are built in, dull black painted plywood walls, about 20 mm thick, being used.

Other appliances are placed on, or fixed to a support of dull black painted plywood having a thickness of 20 mm.

11.7 Replacement

The appliances are operated until steady conditions are established.

11.8 Addition

For appliances intended to be used with a connector incorporating a thermostat, a higher limit is allowed for the temperature rise of the pins of the appliance inlet, provided the appliance connector complies with IS 3010 (Part 1) : 1965 as far as it reasonably applies; the value of the temperature prescribed for the test of clause **10.3.11** of IS 3010 (Part 1) : 1965 and **6.2** of IS 3010 (Part 2) : 1965 is however increased to that attained by the pins of the appliance inlet when the appliance is operated in accordance with the conditions specified in sub-clauses **11.4** and **11.7**.

12 OPERATION UNDER OVERLOAD CONDITIONS OF APPLIANCES WITH HEATING ELEMENTS

This clause of IS 302-1 (1979) is applicable.

13 ELECTRICAL INSULATION AND LEAKAGE CURRENT AT OPERATING TEMPERATURE

This clause of IS 302-1 (1979) is applicable except as follows:

13.2 Addition

For particularized warming plates delivered with or intended to be used with metal vessels the vessels are placed on the heated surface and connected to accessible metal parts, no metal foil being in contact with the accessible heated surface.

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For other appliances, the metal foil is in contact with accessible surfaces of insulating material, no vessels or other objects being placed on the heated surface.

14 RADIO AND TELEVISION INTERFERENCE SUPPRESSION

This clause of IS 302-1 (1979) is applicable.

15 MOISTURE RESISTANCE

This clause of IS 302-1 (1979) is applicable except as follows:

15.2 Addition

Appliances intended to be partially or completely immersed in water the cleaning are subjected to the tests of sub-clause **15.101**.

15.3 Replacement

Instead of the requirements, the following applies:

Appliances where a pan is intended to be placed on the heated surface shall be so constructed that spillage of liquid does not affect their electrical insulation.

Instead of the test procedure specifying how to fill the liquid container, the following applies:

A quantity of water containing approximately 1 percent NaCl, equal to 10 cm³ per 100 cm² of the heated surface is poured steadily over the surface for a period of 1 min.

Additional sub-clause

15.101 Appliances intended to be partially or completely immersed in water for cleaning shall have adequate protection against the effects of immersion.

Compliance is checked by the following tests, which are made on three additional samples.

The samples are operated in accordance with conditions of adequate heat discharge with any thermostat adjusted to the highest setting and at a supply voltage such that the input is 1.15 times rated input, until the thermostat operates for the first time or, for appliances without a thermostat, until steady conditions are established.

The connectors are then withdrawn or the supply otherwise switched off and the samples immediately immersed completely in water having a temperature between 10°C and 25°C.

After 1 h of immersion, the samples are removed from the water and dried care being taken to ensure that all moisture is removed from the insulation in the vicinity of the pins of appliance inlets. The leakage current is then measured as described in sub-clause **16.2**.

For each of the three samples, the leakage current shall not exceed the value specified in sub-clause **16.2**.

The treatment described above and the measurement of the leakage current are made five times, after which the samples shall withstand an electric strength test as specified in sub-clause **16.4**, the test voltage being, however, reduced to 1 000 V.

The sample showing the largest leakage current after the fifth immersion is dismantled, and inspection shall show that water has not entered the appliance to any appreciable extent; in particular, there shall be no trace of water on insulation which could result in a reduction of creepage distances and clearances below the values specified in sub-clause **29.1**.

The remaining two samples are then operated in accordance with conditions of adequate heat discharge for 10 days (240 h).

During this period, the samples are allowed to cool to approximately room temperature five times at regular intervals.

After this period, the connectors of the two samples are withdrawn or the supply otherwise switched off and the samples immediately immersed once more in water for 1 h as described before. They are then dried and the leakage current is measured again as described in sub-clause **16.2**.

For each of the two samples, the leakage current shall not exceed the value specified in sub-clause **16.2**.

Three samples shall then withstand an electric strength test as specified before, and inspection shall show that water has not entered the appliance to any appreciable extent; in particular, there shall be no trace of water on insulation which could result in a reduction of creepage distances and clearance below the values specified in sub-clause **29.1**.

When inspecting the appliances for the response of water, special attention is paid to parts of the appliance in which electrical components are situated.

16 INSULATION RESISTANCE AND ELECTRIC STRENGTH (AFTER HUMIDITY TREATMENT)

This clause of IS 302-1 (1979) is applicable except as follows:

16.2 Addition

For appliances intended to be used with a connector incorporating a thermostat and for appliances intended to be partially or

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completely immersed in water for cleaning, the appliance inlet may be dried, for example, by means of blotting paper, before applying the test voltage if the appliance would not otherwise withstand the test.

17 OVERLOAD PROTECTION

This clause of IS 302-1 (1979) is applicable.

18 ENDURANCE (under consideration)

19 ABNORMAL OPERATION

This clause of IS 302-1 (1979) is applicable except as follows:

19.1 Replacement

Appliances shall be so designed that the risk of fire, mechanical damage impairing safety or the protection against electric shock as a result of abnormal or careless operation is obviated as far as is practicable.

Compliance is checked by the test of 19.2 and if necessary by the test of 19.3 followed by the test of 19.4 for appliances provide with a control which limits the temperature during the test of 11 and by the test of 19.101 for all appliances.

If, in any of the tests, a non-self-resetting thermal cut-out operates, a heating element ruptures or if the current is otherwise interrupted before steady conditions are established the heating period is considered to be ended but if the interruption is due to the rupture of a heating element or of an intentionally weak part the relevant test is repeated on a second sample which shall then also comply with the conditions specified in 19.11.

Rupture of a heating element or of an intentionally weak part in the second sample will not in itself entail a rejection.

As intentionally weak part is a part designed to fail under specimens of abnormal operation so as to prevent the occurrence of a condition which is unsafe within the meaning of this standard. Such a part may be a replaceable component, such as a resistor, a capacitor or a thermal fuse, or a part of a component to be replaced.

Sub-clause 19.1 applies to all appliances

Fuses, thermal cut-outs, overcurrent releases or the like incorporated into the appliance may be used to provide the necessary protection.

If more than one of the tests are applicable for the same appliance these tests are made consecutively.

Additional Sub-clause

19.101 The appliances are operated for 7 h at a supply voltage such that the input is equal to rated input, without a vessel or other object on the heated surface, this surface being however, completely covered with a double-layer of cotton gauge as specified in Annex AA and with a hair-felt pad having a thickness of 25 mm.

If a thermostat starts to cycle, the test is repeated with only one-third of the heated surface furthest from the temperature sensing element covered.

During the test the appliance shall not emit flame or molten metal and there shall be no glowing or flaming of combustible material.

If, after the test, the appliance appears to be usable, it shall withstand the test of sub-clause 16.2

For appliances provided with heating elements with appreciable positive temperature coefficient of resistance the supply voltage is determined as specified in sub-clause 4.1.2

20 STABILITY AND MECHANICAL HAZARDS

This clause of IS 302-1 (1979) is applicable.

21 MECHANICAL STRENGTH

This clause of IS 302-1 (1979) is applicable except as follows:

21.1 Addition

For appliances having surfaces of glass ceramic or similar brittle material in or on which a heating element is mounted or which are essential parts of the enclosure of live parts, three blows are also applied to parts of such surfaces which are, because of the design, not exposed to impacts during the test of 22.101, the hammer spring being however adjusted so that the product of the compression, in millimetres, and the force exerted, in newtons, equals 1 400, the compression being approximately 24 mm. With this adjustment the impact energy is $0.70 = 0.05 \text{ Nm}$.

22 CONSTRUCTION

This clause of IS 302-1 (1979) is applicable except as follows:

Additional Sub-clause

22.101 Appliances having surfaces of glass ceramic or similar brittle material in or on which a heating element is mounted, or which are essential parts of the enclosure of live parts

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shall withstand stresses liable to occur in normal use.

Compliance is checked by the following tests.

A vessel having a bottom of copper or aluminium which is flat over a distance of 12 ± 1 cm and has edge rounded with a radius of at least 1 cm, uniformly filled with sand or shot to a total mass of 1.8 kg is dropped flat from a height of 15 cm onto the surface of glass ceramic or similar brittle material.

This operation is carried out ten times in succession on the cold appliance.

The appliance is then operated at rated voltage or at the upper limit of the rated voltage range until steady conditions are established.

Immediately afterwards, a cotton sheet having dimensions of 40 cm $\times$ 40 cm and a mass between 140 g/m² and 175 g/m² in the dry condition which is soaked with cold water and then folded four times so as to form a pad having dimensions of approximately 10 cm $\times$ 10 cm is applied to the most unfavourable part of the surface of glass ceramic or similar brittle material.

After these tests the surface shall not be broken and shall show no crack visible to the naked eye and the appliance shall withstand the test of sub-clause 16.2.

23 INTERNAL WIRING

This clause of IS 302-1 (1979) is applicable.

24 COMPONENTS

This clause of IS 302-1 (1979) is applicable except as follows:

24.1 Addition

Appliance couplers incorporating thermostats, thermal cut-outs or fuses in the connectors and having dimensions other than those specified in IS 3010 (Part 1) : 1965 and IS 3010 (Part 2) : 1965 shall comply with IS 3010 (Part 1) : 1965 and IS 3010 (Part 2) : 1965 except that:

- the earthing contact of the connector is allowed to be accessible, provided that this contact is not likely to be gripped during insertion or withdrawal of the connector [see sub-clause 10.3.7 of IS 3010 (Part 1) : 1965 and 6.1 of IS 3010 (Part 2) : 1965].
- the breaking-capacity test is carried out using the inlet on the appliance [see sub-clause 10.3.12 of IS 3010 (Part 1) : 1965].

- the temperature rise of current-carrying parts is not determined [see sub-clause 10.3.11 of IS 3010 (Part 1) : 1965 and 6.2 of IS 3010 (Part 1) : 1965].

24.2 Addition

Switches incorporated in the appliances are not required to be switches for frequent operation.

25 SUPPLY CONNECTION AND EXTERNAL FLEXIBLE CABLES AND CORDS

This clause of IS 302-1 (1979) is applicable except as follows:

25.2 Addition

Appliances provided with an appliance inlet other than those standardized in IS 3010 (Part 1) : 1965 and IS 3010 (Part 2) : 1965 shall be delivered with a cord assembly incorporating an appropriate connector.

25.3 Addition

Appliance inlets shall have a rated current of at least 6 A.

25.6 Replacement

Instead of the types of power supply cord specified, the following applies:

Power supply cords shall be not lighter than:

- If rubber insulated, braided flexible cord according to IS 9968 (Part 1) : 1988.
- If polyvinyl chloride insulated, light polyvinyl chloride sheathed flexible cord according to 694 : 1990.

The length of the supply cord measured between the entry point into the appliance and the entry point into the plug and cord guard included, shall be not less than 2.0 m if supplies with a plug or 20 shall not be less than 2.05 m if supplied without plug.

25.11 Addition

for type M attachment, knots in the cord are not used.

A knot in the power supply cord is allowed to serve as a cord anchorage for type Y attachment.

26 TERMINALS FOR EXTERNAL CONDUCTORS

This clause of IS 302-1 (1979) is applicable.

27 PROVISION FOR EARTHING

This clause of IS 302-1 (1979) is applicable.

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28 SCREWS AND CONNECTIONS

This clause of IS 302-1 (1979) is applicable.

29 CREEPAGE DISTANCE AND CLEARANCES

This clause of IS 302-1 (1979) is applicable.

30 RESISTANCE TO HEAT, FIRE AND TRACKING

This clause of IS 302-1 (1979) is applicable.

31 RESISTANCE TO RUSTING

This clause of IS 302-1 (1979) is applicable.

32 RADIATION HAZARDS

This clause of IS 302-1 (1979) is applicable.

101 TESTS

101.1 Type Tests

The tests specified in Table 101 shall constitute the type tests and shall be carried out on two samples of the same type and rating selected preferably at random from a regular production lot. Before commencement of the tests, the warming plates shall be visually examined and inspected for obvious visual defects in respect of components, parts and their assembly, construction, mechanical hazards, markings, provision of suitable terminals for supply connections, earthing and the effectiveness of

Table 101 Schedule of Type Tests
(Clause 101.1)

Sl No.	Tests	Clause Reference
i)	Protection against electric shock	8
ii)	Starting of motoroperated appliances	9
iii)	Input	10
iv)	Temperature rise	11
v)	Operation under overload conditions	12
vi)	Electrical insulation and leakage current at operating temperature	13
vii)	Radio and television interference suppression	14
viii)	Moisture resistance	15
ix)	Insulation resistance and electric strength (after humidity treatment)	16
x)	Overload protection	17
xi)	Endurance (under consideration)	18
xii)	Abnormal operation	19
xiii)	Stability and mechanical hazards	20
xiv)	Mechanical strength	21
xv)	Construction	22
xvi)	Internal wiring	23
xvii)	Components	24
xviii)	Supply connections and external flexible cords	25
xix)	Terminals for external conductors	26
xx)	Provision for earthing	27
xxi)	Screws and connections	28
xxii)	Creepage distance and clearances	29
xxiii)	Resistance to heat fire and tracking	30
xxiv)	Resistance to rusting	31
xxv)	Radiation hazards	32

screws and connections. The external surface finish shall be even and free from finishing defects.

NOTE — See also 4.2.

101.1.1 Criteria of Acceptance

Both samples shall successfully pass all the type tests for proving conformity with the requirements of the standard. If any of the samples fails in any of the type tests, the testing authority at its discretion may call for fresh samples not exceeding twice the original number and subject them again to all tests or to the test(s) in which failure(s) had occurred. No failure should be permitted in the repeat test(s).

101.2 Acceptance Tests

The following shall constitute the acceptance tests:

Test	Clause Reference
a) Protection against electric shock	8
b) Input	10
c) Temperature rise	11
d) Electrical insulation and leakage current at operating temperature	13
e) Moisture resistance	15
f) Insulation resistance and electric strength (after humidity treatment)	16
g) Provisions for earthing	27

NOTE — For the purpose of acceptance tests, the humidity treatment shall be done for 24 hours while conducting the test for moisture resistance (15).

101.2.1 A recommended sampling procedure for acceptance tests is given in Appendix B of IS 302-1 (1979).

101.3 Routine Test

The following shall constitute the routine tests:

Test	Clause Reference
a) Protection against electric shock	8
b) High voltage	13.3.2 of IS 302-1 (1979)
c) Provision for earthing	27

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ANNEX A
TABLE OF TYPE TEST

The Appendix A of IS 302-1 (1979) is not applicable.

ANNEX B
SAMPLING PROCEDURE FOR ACCEPTABLE TEST

The Appendix B of IS 302-1 (1979) is applicable.

ANNEX C
ELECTRONIC CIRCUITS

The Appendix C of IS 302-1 (1979) is applicable if electronic circuits are used.

ANNEX D
MEASUREMENT OF TEMPERATURE WITH THERMOMETER

The Appendix D of IS 302-1 (1979) is applicable.

ANNEX E
ALTERNATIVE TESTS FOR PROTECTED MOTOR UNITS

The Appendix E of IS 302-1 (1979) is applicable.

ANNEX F
IMPACT TEST APPARATUS

The Appendix F of IS 302-1 (1979) is applicable.

ANNEX G
THERMAL CONTROLS AND OVERLOAD RELEASES

The Appendix G of IS 302-1 (1979) is applicable.

ANNEX H
MEASUREMENT OF CREEPAGE DISTANCES AND CLEARANCES

The Appendix H of IS 302-1 (1979) is applicable.

ANNEX J
TEST FOR FIRE-RESISTANCE PROPERTIES

The Appendix J of IS 302-1 (1979) is applicable.

ANNEX K
**BNF JET TEST FOR DETERMINATION OF THICKNESS OF COPPER AND
NICKEL PLATING**

The Appendix K of IS 302-1 (1979) is not applicable.

ANNEX L
**APPROXIMATE MEASUREMENT OF THICKNESS OF CHROMIUM OF NICKEL,
STEEL AND COPPER**

The Appendix L of IS 302-1 (1979) is not applicable.

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ANNEX AA

(Foreword)

SPECIFICATION FOR COTTON GAUZE

The cotton gauze to be used for the test of sub-clause 19.101 is composed as follows:

Material.....	all cotton
Specific mass of gauge.....	approximately 40 g/m ²
Specific mass of threads.....	approximately 20 g/km
Threads per centimeter:	
— warp.....	13
— weft.....	11
Finish.....	bleached

ANNEX BB

(Foreword)

LIST OF REFERRED INDIAN STANDARDS

IS No.	Title	IS No.	Title
302-1 (1979)	Safety of household and similar electrical appliances : Part 1 General requirement (<i>fifth revision</i>)		to and including 1 100 V (<i>first revision</i>)
694 : 1990	PVC insulated cables for working voltage up to and including 1 100 volts (<i>third revision</i>)	3010 (Part 1) : 1965	Appliance connectors and appliance inlet (non reversible three pin type) : Part 1 Appliance connectors
9968 (Part 1) : 1988	Elastomer insulated cables : Part 1 For working voltage up	3010 (Part 2) : 1965	Appliance connectors and appliance inlet (non reversible three pin type) : Part 2 Appliance inlet

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Amendments are issued to standards as the need arises on the basis of comments. Standards are also reviewed periodically; a standard along with amendments is reaffirmed when such review indicates that no changes are needed; if the review indicates that changes are needed, it is taken up for revision. Users of Indian Standards should ascertain that they are in possession of the latest amendments or edition by referring to the latest issue of 'BIS Handbook' and 'Standards Monthly Additions'. Comments on this Indian Standard may be sent to BIS giving the following reference:

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